Foundations of the finite-difference solution of the Black-Scholes PDE

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May 2026; revised October 2026

1 Scope

This note sets up the mathematical and numerical scaffolding on which all the finite-difference and lattice notes rest. It contains no pricer. Its deliverables are:

1. the logarithmic transformation that turns the Black–Scholes PDE into a constant-coefficient parabolic PDE;
2. the truncation of the spatial domain and the choice of artificial Dirichlet boundary conditions;
3. the discretisation grid and the standard centred finite-difference stencils;
4. the von Neumann stability analysis applied to the explicit (FTCS) and implicit (BTCS) schemes for the pure-diffusive part of the PDE.

The pricers themselves (FTCS, BTCS, Crank–Nicolson, PSOR, trinomial) appear in separate notes, all built on this foundation.

2 The Black–Scholes PDE

Under the risk-neutral measure $\mathbb{Q}$, the price $V(S,t)$ of a European option on a non-dividend-paying asset S_t following geometric Brownian motion

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}$$

satisfies the Black–Scholes partial differential equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV \;=\; 0, \qquad S > 0,\; t \in [0,T), \tag{1}$$

with terminal condition $V(S,T) = \phi(S)$, where ϕ is the option payoff. This is a backward parabolic equation in calendar time.

Direct discretisation of (1) on a uniform grid in S is unsatisfactory for two independent reasons. First, the coefficients $\frac{1}{2}\sigma^2 S^2$ and rS depend on the grid position, which complicates the von Neumann stability analysis: the standard analysis assumes constant coefficients. Second, a uniform grid in S resolves the lognormal-tailed distribution of S_T poorly, with too few nodes near $S = 0$ relative to the scale on which $V(S,t)$ varies.

Both problems are removed by a single change of variables.

3 Logarithmic transformation

Introduce

$$x \;=\; \ln\!\left(\frac{S}{K}\right), \qquad \tau \;=\; T - t. \tag{2}$$

The variable x is the log-moneyness; $\tau \in [0, T]$ is the time to maturity. Anchoring at the strike K (rather than using $\ln S$ directly) places $x = 0$ at the at-the-money point, which gives a canonical centring for the truncated domain.

Computing the derivatives by the chain rule, with $S = Ke^x$,

$$\frac{\partial V}{\partial t} \;=\; -\frac{\partial V}{\partial \tau}, \qquad \frac{\partial V}{\partial S} \;=\; \frac{1}{S}\frac{\partial V}{\partial x}, \qquad \frac{\partial^2 V}{\partial S^2} \;=\; \frac{1}{S^2}\left(\frac{\partial^2 V}{\partial x^2} - \frac{\partial V}{\partial x}\right).$$

Substituting into (1) and rearranging yields

$$\boxed{\frac{\partial V}{\partial \tau} \;=\; \frac{\sigma^2}{2}\frac{\partial^2 V}{\partial x^2} + \mu\,\frac{\partial V}{\partial x} - r\,V, \qquad \mu \;:=\; r - \frac{\sigma^2}{2}.} \tag{3}$$

This is a forward parabolic equation in τ on the unbounded domain $x \in \mathbb{R}$, with constant coefficients. The drift μ in (3) is precisely the drift of $\ln S_t$ under $\mathbb{Q}$, by Itô.

The initial condition becomes

$$V(x, 0) \;=\; \phi(Ke^x), \qquad x \in \mathbb{R}.$$

Remark 3.1. *The transformation* (2) *preserves the regularity of V. The non-smoothness of ϕ at the strike (the kink of $(\cdot)^+$) becomes a kink of the transformed initial condition at $x = 0$. This irregularity is the source of the spurious oscillations of Crank–Nicolson discussed in the Crank–Nicolson note.*

4 Structure of the transformed PDE

Equation (3) is a constant-coefficient *convection–diffusion–reaction* (CDR) equation:

$$V_\tau \;=\; D\,V_{xx} \,+\, a\,V_x \,+\, c\,V, \qquad D = \frac{\sigma^2}{2} > 0, \quad a = \mu \in \mathbb{R}, \quad c = -r \le 0.$$

Each of the three terms has a distinct numerical character.

- The *diffusive* term $D\,V_{xx}$ is dominant for typical financial parameters ($\sigma \in [0.1, 0.5]$, Δx small). It governs the most restrictive stability constraint on the explicit FTCS scheme.
- The *convective* term $a\,V_x$ encodes directional drift. For $\mu > 0$ information flows toward $+x$; for $\mu < 0$ toward $-x$. With centred space differences (used throughout the finite-difference notes) the contribution to the von Neumann factor is purely imaginary.
- The *reactive* term $cV = -rV$ is local in space and acts as a uniform exponential decay with rate r. In the implicit scheme it adds $r\Delta\tau$ to the denominator of the amplification factor and can only reduce $|g|$. In the explicit scheme it shifts g by $-r\Delta\tau$: this never affects stability under the generalised von Neumann criterion $|g| \le 1 + O(\Delta\tau)$, but under the strict criterion $|g| \le 1$ it slightly tightens the bound on α (Section 9).

Remark 4.1 (Why we keep all three terms together)**.** *A common simplification in textbooks is to apply a further change of variables $V = e^{-r\tau}U$ (kills the reactive term) and a Galilean shift in x (kills the convective term), reducing* (3) *to the pure heat equation $U_\tau = D\,U_{xx}$. This is mathematically clean but obscures the discrete structure: the boundary conditions and the payoff become time-dependent in non-trivial ways, and the relationship to the original V is harder to read off the code. Following the standard practice of Tavella & Randall [1], we discretise* (3) *as it stands.*

5 Truncation of the spatial domain

The unbounded domain $x \in \mathbb{R}$ has to be truncated to a finite interval $[x_{\min}, x_{\max}]$. The truncation introduces an artificial boundary; on it we will impose Dirichlet conditions whose values are read off the analytic asymptotics of V.

A standard choice is symmetric truncation centred at $x = 0$:

$$x_{\min} \;=\; -N_\sigma\,\sigma\sqrt{T}, \qquad x_{\max} \;=\; +N_\sigma\,\sigma\sqrt{T}, \tag{4}$$

with N_σ a small constant (typically 4 or 5). The justification is that the lognormal density of $S_T = Ke^{x_T}$ has $x_T \sim \mathcal{N}(\mu T,\, \sigma^2 T)$ under $\mathbb{Q}$, so the probability mass outside $[-N_\sigma\sigma\sqrt{T},\, +N_\sigma\sigma\sqrt{T}]$ is $O\big(e^{-N_\sigma^2/2}\big)$, which for $N_\sigma = 4$ is $\approx 6 \times 10^{-5}$ and decays super-polynomially in N_σ.

Remark 5.1. *The total error of any FD pricer is the sum of three contributions: the discretisation error (powers of Δx and $\Delta\tau$), the truncation error from the artificial boundary, and floating-point round-off. For $N_\sigma \geq 4$ the truncation contribution is typically below the discretisation error already at moderate $N \sim 100$ and never becomes the dominant error, so we treat it as negligible throughout. This separation of error sources reappears in the Crank–Nicolson note, where the empirical convergence rate of CN is verified.*

6 Boundary and initial conditions

European call

The terminal payoff $\phi(S) = (S - K)^+$ becomes, in x,

$$V(x, 0) \;=\; K\,(e^x - 1)^+, \qquad x \in [x_{\min}, x_{\max}]. \tag{5}$$

For the boundary conditions we use the well-known asymptotic limits. As $S \to 0$ the call goes to zero; as $S \to \infty$ it tends to the forward $S - Ke^{-r(T-t)}$. Translating to (x, τ),

$$V(x_{\min}, \tau) \;=\; 0, \tag{6}$$

$$V(x_{\max}, \tau) \;=\; K\big(e^{x_{\max}} - e^{-r\tau}\big)\,. \tag{7}$$

Both equalities hold in the limit $x_{\min} \to -\infty$, $x_{\max} \to +\infty$. On a truncated domain they are imposed as exact Dirichlet conditions; the small additional error this introduces is the truncation error discussed above.

European put

By analogous reasoning, the put payoff $\phi(S) = (K - S)^+$ becomes

$$V(x, 0) \;=\; K\,(1 - e^x)^+, \tag{8}$$

and the asymptotic boundary values are

$$V(x_{\min}, \tau) \;=\; K\big(e^{-r\tau} - e^{x_{\min}}\big)\,, \tag{9}$$

$$V(x_{\max}, \tau) \;=\; 0. \tag{10}$$

Equation (9) is a simple consequence of the martingale property: as $S \to 0$, $\mathbb{Q}\{S_T < K\} \to 1$ so $V(S, t) \to e^{-r(T-t)}\big(K - \mathbb{E}^{\mathbb{Q}}[S_T \mid S_t = S]\big) = Ke^{-r(T-t)} - S$, which is precisely $K(e^{-r\tau} - e^x)$.

Remark 6.1. *For the American put the boundary conditions* (9)–(10) *have to be replaced by the* intrinsic-value *conditions $V(x_{\min}, \tau) = K - Ke^{x_{\min}}$ (the option is exercised immediately for very low S) and the same $V(x_{\max}, \tau) = 0$. This is treated in the note on the American put via PSOR.*

7 The discretisation grid

We discretise on a uniform tensor-product grid in (x, τ). Let

$$N \in \mathbb{N}, \quad M \in \mathbb{N}, \qquad \Delta x \;=\; \frac{x_{\max} - x_{\min}}{N}, \qquad \Delta\tau \;=\; \frac{T}{M}. \tag{11}$$

The spatial nodes are $x_j = x_{\min} + j\,\Delta x,\ j = 0, 1, \ldots, N$, giving $N+1$ nodes; the time levels are $\tau_n = n\,\Delta\tau,\ n = 0, 1, \ldots, M$, giving $M+1$ levels.

The discrete unknown is $V_j^n \approx V(x_j, \tau_n)$. The *interior nodes* are $j = 1, \ldots, N-1$ (a total of $N-1$ unknowns at each time level); the boundary nodes $j = 0$ and $j = N$ are fixed by the Dirichlet conditions of Section 6. At $\tau_0 = 0$ all nodes are fixed by the initial condition; the answer of interest is $V(0, T)$ if $S_0 = K$, or more generally the interpolant of V_j^M at $x = \ln(S_0/K)$.

The uniformity of the grid in x (not in S) is the consequence of the logarithmic transformation: a uniform grid in S would under-resolve the region near $S = 0$ and over-resolve large S, since V varies on a scale proportional to S (of order $\sigma S\sqrt{T}$). The transformation makes the natural sampling uniform in log-moneyness, which is also the natural scale for V to vary on.

8 Finite-difference stencils

Let f be a smooth function. Taylor expansion gives

$$f(x \pm h) \;=\; f(x) \pm h f'(x) + \frac{h^2}{2} f''(x) \pm \frac{h^3}{6} f'''(x) + O(h^4).$$

Combining the $+h$ and $-h$ expansions yields the centred stencils

$$f'(x) \;\approx\; \frac{f(x+h) - f(x-h)}{2h} + O(h^2), \tag{12}$$

$$f''(x) \;\approx\; \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2). \tag{13}$$

For the time derivative we use forward or backward Euler:

$$\partial_\tau V|_j^n \;\approx\; \frac{V_j^{n+1} - V_j^n}{\Delta\tau} + O(\Delta\tau), \qquad \text{(forward, used in FTCS)} \tag{14}$$

$$\partial_\tau V|_j^n \;\approx\; \frac{V_j^n - V_j^{n-1}}{\Delta\tau} + O(\Delta\tau). \qquad \text{(backward, used in BTCS)} \tag{15}$$

Centred-in-time differences (used in Crank–Nicolson) achieve $O(\Delta\tau^2)$ at the cost of being implicit at both adjacent levels.

The label *FTCS* stands for *Forward-Time, Centred-Space* (combining (14) with (12)–(13) evaluated at level n); *BTCS* for *Backward-Time, Centred-Space* (combining (15) with the same spatial stencils, but evaluated at level $n+1$ from the perspective of the unknown).

9 Von Neumann stability analysis

Stability of a finite-difference scheme is the property that round-off errors do not grow unboundedly as the time-stepping proceeds. Together with consistency (the truncation error tends to zero as $\Delta x, \Delta\tau \to 0$), it implies convergence:

Theorem 9.1 (Lax equivalence, [4])**.** *For a well-posed linear initial-value problem and a consistent finite-difference approximation, stability is necessary and sufficient for convergence.*

The criterion of stability we will use is the von Neumann criterion, which exploits the fact that on a uniform grid with constant coefficients the discrete Fourier modes $e^{ij\theta}$, $\theta \in [-\pi, \pi]$, diagonalise the spatial operators.

9.1 Setup

Write the round-off error at the level n as a single Fourier mode,

$$\varepsilon_j^n \;=\; g(\theta)^n\, e^{ij\theta}, \tag{16}$$

with $g(\theta) \in \mathbb{C}$ the (mode-dependent) *amplification factor*. Linearity of the scheme allows independent treatment of each θ. The scheme is *von Neumann stable* iff

$$|g(\theta)| \;\le\; 1 \qquad \text{for all } \theta \in [-\pi, \pi]. \tag{17}$$

9.2 FTCS for the pure-diffusive equation

Consider first the model problem $V_\tau = DV_{xx}$, ignoring the convective and reactive terms; we will reinstate them below. The FTCS discretisation is

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; D\,\frac{V_{j+1}^n - 2V_j^n + V_{j-1}^n}{\Delta x^2}.$$

Substituting the ansatz (16), dividing by $g^n e^{ij\theta}$, and using $e^{i\theta} + e^{-i\theta} = 2\cos\theta$:

$$g - 1 \;=\; \frac{D\,\Delta\tau}{\Delta x^2}\left(2\cos\theta - 2\right) \;=\; -4\,\alpha\,\sin^2\tfrac{\theta}{2},$$

where the dimensionless *Fourier number* is

$$\alpha \;:=\; D\,\frac{\Delta\tau}{\Delta x^2} \;=\; \frac{\sigma^2}{2}\cdot\frac{\Delta\tau}{\Delta x^2}. \tag{18}$$

Therefore

$$g(\theta) \;=\; 1 - 4\alpha\,\sin^2\tfrac{\theta}{2} \;\in\; \mathbb{R}.$$

The criterion (17) becomes $-1 \le g \le 1$. The upper bound is automatic since $\alpha, \sin^2 \ge 0$. The lower bound is tight at $\theta = \pi$ and gives

$$\boxed{\alpha \le \tfrac{1}{2}} \quad\Longleftrightarrow\quad \frac{\sigma^2\Delta\tau}{\Delta x^2} \le 1. \tag{19}$$

(19) is the *CFL condition* for the heat equation, in honour of Courant, Friedrichs and Lewy. It says that an explicit diffusive scheme cannot propagate the heat by more than half a grid spacing per time step. The constant $1/2$ is sharp.

Remark 9.2. *The CFL constraint is severe in practice. For $\sigma = 0.2$, $\Delta x = 0.05$, the largest stable $\Delta\tau$ is $\alpha\Delta x^2/D \approx \frac{0.5\cdot 0.0025}{0.02} = 0.0625$, so a year requires at least $T/\Delta\tau = 16$ time steps. Halving Δx requires* quadrupling *the number of time steps. This $O(\Delta x^{-2})$ scaling of the time-step count is the principal limitation of FTCS and the motivation for the implicit schemes (BTCS, Crank–Nicolson).*

9.3 BTCS for the pure-diffusive equation

The implicit scheme evaluates the spatial stencil at level $n+1$:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; D\,\frac{V_{j+1}^{n+1} - 2V_j^{n+1} + V_{j-1}^{n+1}}{\Delta x^2}.$$

Substituting (16) gives

$$g - 1 \;=\; -4\alpha\, g\,\sin^2\tfrac{\theta}{2} \;\Longrightarrow\; g(\theta) \;=\; \frac{1}{1 + 4\alpha\sin^2(\theta/2)}.$$

Since $\alpha > 0$, the denominator is at least 1 for every θ, so $g \in (0, 1]$. The criterion is satisfied for any $\alpha > 0$:

Proposition 9.3 (Unconditional stability of BTCS). *The BTCS scheme for $V_\tau = DV_{xx}$ is von Neumann stable for any $\Delta\tau > 0,\ \Delta x > 0$.*

This is the central trade-off of the finite-difference methods. FTCS is cheap per step (matrix-free) but has the CFL constraint; BTCS allows arbitrary $\Delta\tau$ but requires solving a linear system at each step. The BTCS note pays the cost of that linear solve via the Thomas algorithm.

9.4 Convective term: an aside

Reinstating the convective term aV_x with centred-space differences in FTCS adds $a\,\Delta\tau\,(V^n_{j+1} - V^n_{j-1})/(2\Delta x)$ to the update; since $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$, the amplification factor becomes complex,

$$g(\theta) \;=\; 1 - 4\alpha\sin^2\tfrac{\theta}{2} + 2i\,\nu\,\sin\theta, \qquad \nu \;:=\; a\,\frac{\Delta\tau}{2\,\Delta x},$$

where ν is the *Courant number* (advective). Imposing $|g|^2 \le 1$ yields a mixed condition coupling α and ν. With $s := \sin^2(\theta/2) \in [0,1]$ and $\sin^2\theta = 4s(1-s)$,

$$|g|^2 - 1 \;=\; 8s\,\big[\,2(\alpha^2 - \nu^2)\,s + (2\nu^2 - \alpha)\,\big],$$

and the bracket, linear in s, is non-positive on $(0,1]$ iff it is non-positive at $s = 0$ and at $s = 1$ (see [4, Ch. 2 and 6]):

$$2\nu^2 \;\le\; \alpha \quad \text{and} \quad \alpha \le \tfrac{1}{2}.$$

In the variables of (3) the first condition reads $\mu^2\Delta\tau \le \sigma^2$. For Black–Scholes parameters the relevant figure of merit is the *grid Péclet number* $\mathrm{Pe}_h = |a|\Delta x/(2D) = |\mu|\Delta x/\sigma^2$. For $\sigma = 0.2$, $|\mu| \le 0.1$, $\Delta x \le 0.1$, $\mathrm{Pe}_h \le 0.25$, so the convective contribution is sub-dominant and the diffusive bound (19) governs in practice. The reactive term $-rV$, being local in space, shifts g by $-r\Delta\tau$, which increases $|g|$ wherever $\mathrm{Re}\,g < 0$. At $\theta = \pi$, where $g = 1 - 4\alpha - r\Delta\tau$, the strict criterion (17) therefore requires $\alpha \le \frac{1}{2} - \frac{r\Delta\tau}{4}$. Under the generalised von Neumann criterion $|g| \le 1 + K\Delta\tau$, with K independent of the grid (the notion of stability that the Lax equivalence theorem requires), an $O(\Delta\tau)$ shift never affects stability and the bound remains $\alpha \le 1/2$.

9.5 Summary: two control numbers

The discrete dynamics of all these schemes is controlled by two dimensionless numbers,

$$\alpha \;=\; \frac{\sigma^2}{2}\cdot\frac{\Delta\tau}{\Delta x^2}, \qquad \nu \;=\; \frac{|\mu|}{2}\cdot\frac{\Delta\tau}{\Delta x}. \tag{20}$$

FTCS is stable iff $\alpha \le 1/2$ and the convective contribution does not violate the mixed bound; in the parameter range of this project this reduces effectively to $\alpha \le 1/2$. BTCS and Crank–Nicolson are unconditionally stable in α. As shown in the trinomial-tree note, the Kamrad–Ritchken trinomial tree maps to FTCS with $\alpha = 1/(2\lambda)$; its non-negative-probability condition $p_m = 1 - 1/\lambda \ge 0$, that is $\lambda \ge 1$ or $\alpha \le 1/2$, is the discrete-tree avatar of the CFL constraint (19).

10 Implementation outline

The implementation (`quantlib.pde`) provides five ingredients:

1. `Grid` (Python dataclass / C++ struct): an immutable container holding N, M, T, σ, r, K, $x_{\min}$, $x_{\max}$, Δx, $\Delta\tau$, the arrays $\{x_j\}$ and $\{\tau_n\}$.
2. `build_grid`: a constructor that picks $x_{\min}, x_{\max}$ from $\pm N_\sigma\sigma\sqrt{T}$ (default $N_\sigma = 4$) and assembles the grid.

3. Initial-condition functions `call_initial_condition`, `put_initial_condition` returning the payoff vector at $\tau = 0$.

4. Boundary functions `call_boundary_lower`, `call_boundary_upper`, `put_boundary_lower`, `put_boundary_upper` returning Dirichlet values for every τ_n.

5. Stability-number utilities `fourier_number` and `courant_number` for diagnostic use by the explicit FTCS pricer.

This module imports nothing from outside `numpy` and the standard library. It contains no pricer.

References

References

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